

Uzman Ali

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About me

Antenna Design Engineer with 6+ years of experience designing, simulating, prototyping, testing, and optimizing antennas for wireless communication, IoT, RFID, GNSS, Bluetooth, radar, and biomedical applications. Proficient in CST Studio Suite, ANSYS HFSS, RF measurements, antenna integration, EMC analysis, impedance matching, and product validation. Experienced collaborating with RF, hardware, and mechanical engineers to deliver antenna solutions from concept through laboratory and field verification. Strong background in technical documentation, troubleshooting, and R&D-driven product development.

Projects

Antenna Platforms

Designed, simulated, optimized, and characterized broadband and multiband antennas using CST Studio Suite and ANSYS HFSS, including impedance matching, radiation pattern optimization, SAR analysis, and wireless link validation

Industrial RF Simulation and Design Consulting

Supported industrial customers using ANSYS HFSS for antenna design, RF system optimization, EMC analysis, and product development

Implant-to-Implant RF Communication System

- Designed and experimentally validated a complete wireless communication platform for implantable devices
- Developed antennas, measurement setups, wireless link evaluation methodologies, and system-level performance validation

Other Industrial Projects:

- Multi-radio LTE/Bluetooth/GNSS antenna development for connected vehicle IoT platforms
- RFID reader antenna design and validation for inventory management and asset tracking systems
- Bluetooth antenna development for embedded IoT sensor nodes operating within concrete infrastructure
- 120 GHz radar array antenna design, simulation, and optimization for high-resolution sensing applications
- RFID tag antenna development for anti-counterfeiting and product authentication of bottled goods
- Bluetooth antenna design for wireless smoke detection and monitoring systems
- Horn antenna design and characterization for biomedical and implant communication research
- ISM-band implantable capsule antenna design, simulation, and experimental validation

Work experience

Doctoral Researcher | Tampere University | 24/04/2023 - Current | Tampere, Finland

- Conducted research on RF systems, antenna technologies, electromagnetic propagation, and wireless communications, combining simulation, prototyping, and experimental validation
- Designed, optimized, and characterized multiband and broadband antenna systems using CSTStudioSuite and ANSYS HFSS and electromagnetic modelling techniques
- Developed and implemented RF measurement methodologies and experimental setups for wireless communication and antenna characterization
- Executed RF testing and verification activities, including VNA measurements, antenna characterization, SAR analysis, and wireless performance assessment
- MATLAB for processing, visualization, and analysis of large electromagnetic and measurement datasets
- Collaborated with multidisciplinary researchers and industry partners to translate research concepts into practical wireless solutions
- Published and presented research results in leading IEEE journals and international conferences

Application Engineer (Hihg Frequency) | EDR & Medeso Oy | 24/01/2022 - 23/04/2023 | Tampere, Finland

- Supported industrial product development programs involving antenna integration, RF system design, electromagnetic compatibility, and wireless performance optimization
- Provided technical consulting, troubleshooting, and workflow optimization for high-frequency simulation and product development activities
- Supported antenna design, RF system analysis, and electromagnetic compatibility assessments through simulation-driven engineering approaches
- Conducted technical demonstrations, customer training sessions, and workshops related to RF and electromagnetic simulation technologies
- Collaborated with R&D teams across telecommunications, industrial electronics, and medical technology

Antenna Design Engineer | Radientum Oy | 01/04/2020 - 16/07/2021 | Tampere, Finland

- Designed and optimized antennas for commercial wireless products from concept through prototype validation
- Supported antenna integration into space-constrained IoT and wireless devices
- Performed RF measurements, matching network optimization, impedance tuning, and performance verification
- Worked closely with multidisciplinary product development teams to balance RF performance, manufacturability, and project requirements
- Participated in prototype development and product validation activities supporting commercial product releases

Education & Training

Doctoral Studies (Thesis Submitted) | 24/04/2023 - Current | Tampere, Finland

Level in EQF: 8 | **Thesis:** Development of Implantable Antennas for Intracranial Applications and Implant-to-Implant Communication

M.Sc. Electrical Engineering with Specialization in Wireless Communication and RF Systems | Tampere University | Tampere, Finland

Thesis: Comparison of Conventional 50-ohm Impedance Matching and Antenna Self Matching for Energy Harvester

Career Breaks

Laid-Off

Following a COVID-19 related layoff, I strengthened my resilience, self-management, and ability to remain focused and productive in high-pressure situations.

Indistrial and Scientific Activities

Technical Leadership & Collaboration

- Supervised master's students in antenna engineering, RF measurements, and scientific publishing
- Collaborated with international researchers on antennas, RF energy harvesting, and wireless power transfer
- Contributed to peer-reviewed publications through international research partnerships
- Mentored bachelor’s students in developing research skills and entering academic research

Scientific Activities

- Presented research at major international conferences (EuCAP, APMC, APS-URSI) and national-level research seminars (RF Sampo, LFTY)
- Engaged in continuous professional development through industrial and academic training events, including Ansys and Simulia workshops, RF related summer schools

References

PhD Supervisor

Toni Björninen

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Manager

Antti-Jussi Romppanen

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